Milestone 3 — Design of Experiments & High-Fidelity Simulation Database (Steps 3–4)

Deliverables: - *"Comprehensive Parameter Space"* (Step 3); - *"Large-Scale Simulation Data"* (Step 4).

Script: `milestone3/run_milestone3.m` (log: `results/data/M3_summary.txt`) **Database:** `results/doe/shm_database.mat`, `design_database.mat`, `*_scalars.csv`

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1. Purpose

The AI of Milestones 5–6 can only learn what the data shows it. Step 3 decides **which wings, environments and damages are simulated**, and Step 4 **simulates them with the validated FSDT solver** and stores the results in a fixed, documented format. Three questions drive the design:

1. **What varies in reality?** Manufacturing scatter, temperature, root fixity, and damage type, location, size and severity.
2. **How do we cover that space efficiently?** Space-filling sampling instead of grids.
3. **What must be stored** so that every later step can be done *without re-running the solver*?

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2. Two-tier database strategy

Tier	Purpose in the digital twin	Samples	Variables
SHM tier	Diagnose *this* wing in service: detect, localise, classify and quantify damage despite environmental variability	**2 800** = 7 classes × 400	18
Design tier	Generalisation: a parametric surrogate / PINN valid across designs; transfer to other wings	**400** healthy designs	11

2.1 Why load magnitude is *not* a DOE variable — superposition

The structural model is linear, K d = F. For any load factor n, the response is d(n) = n · d(1). Sampling n would waste a dimension. Instead, the **unit responses** are stored: - L1: net 1 g flight load (lift minus weight); - L2: 50 N tip test load; - the modal forces $\Gamma = \Phi^T F$, which with the stored frequencies and damping regenerate *any* dynamic response.

Load magnitude and type are reintroduced cheaply in Milestone 4 as random scaling plus sensor noise. This is the “load magnitudes & types” item of Step 3, handled by physics rather than by brute force.

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3. Variables and ranges (SHM tier)

Group	Variable	Range	Physical basis
Material scatter	E_1, E_2, G_{12} (face) scale	±5 %, ±8 %, ±8 %	Typical batch-to-batch CV of prepreg properties (fibre- vs matrix-dominated)
	G_c, E_c (core) scale	±15 %	Foam density scatter
	ρ_f, ρ_c scale	±3 %, ±10 %	Resin content, foam density
Geometry scatter	ply thickness, core thickness	±5 %, ±3 %	Cured-ply-thickness variation, machining tolerance
	global ply misalignment Δθ	±2°	Hand lay-up tolerance
Environment	temperature T	−10 … 40 °C	Operational envelope
Boundary	root fixity k* = k_θ L / D_{11}	200 … 5000 (log)	Root fitting compliance (k* → ∞ = clamp)
Damage	centre x, y	0.03–0.97 L, 0.12–0.88 c	Anywhere on the wing
	radius r, aspect ratio	15–60 mm, 0.6–1.4	Barely-visible impact to large debond
	severity D	0.1 – 0.9	Stiffness loss (1 − D)
	face	top / bottom	Impact side, inside/outside

Damage classes (stratified, 400 each): healthy, core debond, skin matrix cracking, skin fibre breakage, core crush, delamination, root loosening. - Impact-type damages (fibre breakage, core crush) use a Gaussian severity profile; the others are uniform. - Strengths degrade together with stiffness, so the strength result is a **residual strength**.

3.1 Environmental model (temperature)

Below T_g, epoxy-matrix and foam moduli fall roughly linearly with temperature. With reference T_0 = 20 °C:

$$P(T) = P_{20}\,[1 - a\,(T - T_0)], \qquad a_{matrix} = 0.003\,/^{\circ}C, \quad a_{fibre} = 0.0002\,/^{\circ}C, \quad a_{core} = 0.0025\,/^{\circ}C$$

This applies to E_2, G_{12}, G_{13}, G_{23} and the matrix strengths of the faces, to E_1/X of the fibres, and to all core properties. A 30 °C rise lowers f_2 (torsion) by 2.4 %. That is **larger than most damage effects**, and it is the central difficulty of real SHM.

3.2 Boundary-condition model (elastic root)

The clamp is replaced by distributed rotational springs at the root. Translations stay fixed.

$$k_b = k^* \frac{D_{11}}{L}, \qquad k_t = k^* \frac{D_{22}}{L} \qquad \text{per unit root length}$$

k*	30	100	300	1000	3000	∞
Δf_1 [%]	−5.4	−1.7	−0.59	−0.18	−0.06	0

Root loosening (a 7th damage class: loose bolts or a failed bond) multiplies the springs in a chord-wise band by (1 − D).

3.3 Design tier variables

- Four face ply angles, each −90°...90° (symmetric layup preserved).
- Ply thickness 0.10–0.20 mm and core thickness 8–30 mm, i.e. the **thickness ratio** t_f / t_c.
- Face material: T300/5208, AS4/3501-6 or E-glass/epoxy.
- Core grade 0.5–2 (log). Gibson–Ashby foam scaling G ∝ ρ², so ρ_c ∝ √grade.
- Planform taper 0.5–1 and LE sweep 0–20°.
- Root fixity.

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4. Sampling theory

4.1 Latin Hypercube Sampling (McKay, Beckman & Conover 1979)

- Each variable's range is split into N equal-probability strata.
- Each stratum is sampled exactly once, and the strata of different variables are randomly paired.
- Result: perfect one-dimensional coverage for any N, and an estimator variance never worse than Monte-Carlo for monotone responses.

4.2 Maximin optimisation (Morris & Mitchell 1995)

Among Latin hypercubes, prefer the one whose closest pair of points is farthest apart:

$$X^* = \arg \max_X \min_{i \neq j} \| x_i - x_j \|_2$$

Implementation (`src/doe/lhs_maximin.m`, no toolbox required): 1. Take the best of 20 random LHS designs. 2. Apply 400 greedy element exchanges within columns. These preserve the Latin property and are accepted only if the minimum distance increases.

4.3 Uniformity metric — centred L2-discrepancy (Hickernell 1998)

$$CD_2^2 = \left(\tfrac{13}{12}\right)^k - \frac{2}{n}\sum_i \prod_j \left(1 + \frac{|z_{ij}|}{2} - \frac{z_{ij}^2}{2}\right) + \frac{1}{n^2}\sum_{i,l} \prod_j \left(1 + \frac{|z_{ij}|}{2} + \frac{|z_{lj}|}{2} - \frac{|x_{ij} - x_{lj}|}{2}\right), \quad z = x - \tfrac{1}{2}$$

Lower is more uniform.

4.4 Stratification by class

An **independent** maximin LHS is drawn for each damage class. Every class therefore: - has equal size (balanced data for classification); - covers its own location, size and severity space uniformly.

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5. Sensitivity analysis theory

5.1 Morris elementary effects (Morris 1991; Campolongo et al. 2007)

- Run r random one-at-a-time trajectories on a p-level grid. Each trajectory costs k + 1 model runs.
- Each factor's elementary effect is

$$EE_i = \frac{y(\mathbf{x} + \Delta \mathbf{e}_i) - y(\mathbf{x})}{\Delta}, \qquad \mu_i^* = \frac{1}{r}\sum |EE_i|, \qquad \sigma_i = \mathrm{std}(EE_i)$$

- μ* ranks overall influence; σ flags non-linearity or interactions.
- Setup: k = 12 variability factors, r = 20, so 260 runs.

5.2 First-order Sobol indices (variance decomposition)

$$S_i = \frac{\mathrm{Var}_{x_i}(E[y \mid x_i])}{\mathrm{Var}(y)}$$

S_i is estimated from the already computed LHS samples by binning x_i into 10 equal-count bins (`sobol_binned.m`), so no extra runs are needed. Σ S_i ≈ 1 means the response is additive.

5.3 Paired "healthy twin" analysis

To separate the **pure damage effect** from the variability, a subset of damaged samples (60 per class) is re-simulated with identical material, temperature and boundary values but **no damage**. The difference is then caused only by the damage, and it can be compared directly with the healthy variability (EOV std).

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6. What one high-fidelity simulation computes (`src/doe/run_sample.m`)

1. **Model:** perturbed laminate, elastic root (springs scaled locally for root loosening), damaged sections (continuum damage, strengths degraded).
2. **Modal analysis**, 8 modes:
   - frequencies;
   - automatic mode labels (1B, 1T, ...);
   - mass-normalised w-shapes at all 451 nodes;
   - bottom-skin modal strains at all 400 elements.
3. **Static unit cases L1, L2:** deflection fields, top and bottom surface strains (ε_x, ε_y, γ_xy), tip deflection, twist.
4. **Residual strength** at ultimate load (n_ult · L1): Tsai–Wu, Hashin, core shear and interlaminar reserve factors, read at x ≥ 50 mm (Milestone 2 root singularity).
5. **Gust response** by **exact modal superposition** (`modal_response.m`):
   - every mode is integrated exactly for piecewise-linear loading (first-order-hold matrix exponential);
   - plus a static correction for the truncated modes (mode-acceleration method);
   - validated against full Newmark: errors 0.03 % in displacement and 0.04 % in acceleration (V11), and about 1000× faster.

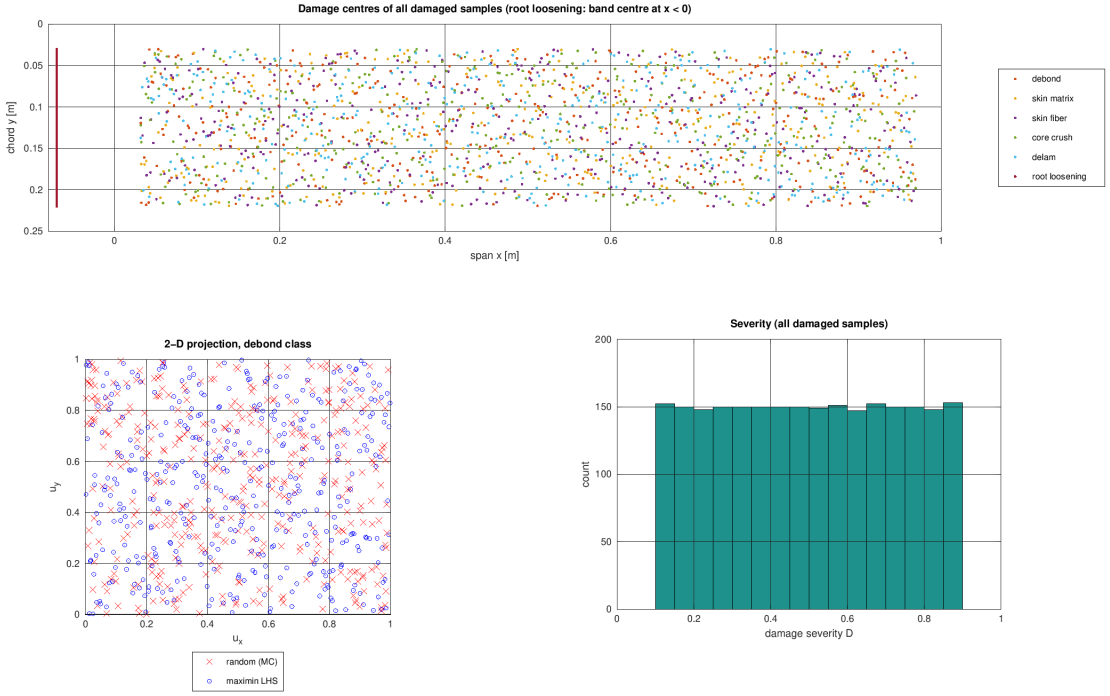
   Two gusts are computed: the design gust and a sharp gust with $t_g = T_1$.
6. **Quality checks:** global equilibrium and eigen-residual $\|K\varphi - \omega^2 M\varphi\|/\|K\varphi\|$ on the free DOFs.

Damping: Rayleigh coefficients are fitted **once** on the nominal wing (ζ = 2 % at modes 1 and 5). Damping is a material and joint property, so a damaged wing with shifted frequencies gets consistently shifted damping ratios.

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7. Results — sample plan quality

	SHM tier	Design tier
Samples	2 800 (7 × 400)	400
Variables	18	11
Min. inter-point distance, maximin-LHS vs random	0.803 vs 0.708 (**+13 %**)	0.452 vs 0.405
Centred L2 discrepancy, LHS vs random	0.312 vs 0.359 (**−13 %**)	0.110 vs 0.141



Coverage

- Damage centres cover the whole planform uniformly for every class.
- Severity is flat over 0.1–0.9, as expected from Latin stratification.
- The 2-D projection shows LHS points spread evenly where random points cluster and leave gaps.

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8. Results — database quality checks

Check	Result
Successful simulations	**2800 / 2800** (SHM), **400 / 400** (design): zero failures

Check	Result
Max. global equilibrium error	1.2×10^{-11}
Max. eigen-residual $\|K\varphi - \omega^2 M\varphi\|/\|K\varphi\|$	2.3×10^{-11}
Run time per sample (Octave, 4 parallel workers)	2.05 s (SHM), 7.05 s (design: tapered meshes, no element cache)
Nominal mode sequence	1B 1T 2B 1I(in-plane) 3B 2T 4B 3T
Mode re-ordering, SHM tier	0.0 % of samples
Mode re-ordering, design tier	**64.8 %** of designs

> **Consequence for Milestone 4.** Across designs, the order of modes changes in two out of three wings: torsion can drop below 2nd bending, an in-plane mode can move, and so on. Features must therefore be matched by **mode type / shape (MAC tracking)**, never by mode index alone. The database stores the automatic mode labels to make this easy.

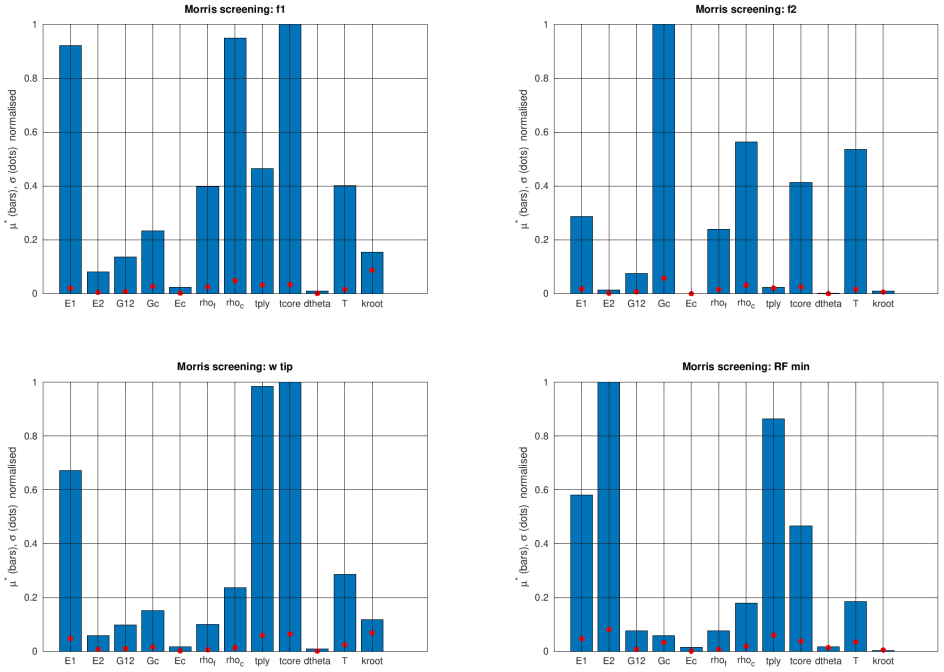
Bug found and fixed during QA. The first design-tier run sampled the foam grade from 3–100× instead of 0.5–2×, because the bounds of a log-scaled variable must be given as $\log_{10}$ exponents. This produced semi-wings of 3.3 kg and negative tip deflections. It was caught by a physical plausibility check. The convention is now documented in `doe_config.m` and the tier was regenerated. The SHM tier was not affected.

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9. Results — sensitivity analysis (healthy wing)

9.1 Morris screening (260 runs), μ* normalised per output

Factor	f_1 (1B)	f_2 (1T)	f_3 (2B)	w_tip	twist	RF_min
E_1	0.92	0.29	0.59	0.67	0.29	0.58
E_2	0.08	0.01	0.05	0.06	0.01	**1.00**
G_{12}	0.14	0.07	0.09	0.10	0.07	0.08
G_c (core shear)	0.23	**1.00**	**1.00**	0.15	**1.00**	0.06
E_c	0.02	0.00	0.01	0.02	0.00	0.01
ρ_f	0.40	0.24	0.36	0.10	0.00	0.08
ρ_c	0.95	0.56	0.85	0.24	0.00	0.18
t_ply	0.47	0.02	0.15	0.98	0.38	0.86
t_core	**1.00**	0.41	0.73	**1.00**	0.60	0.47
Δθ	0.01	0.00	0.01	0.01	0.03	0.02
T	0.40	0.54	0.62	0.29	0.54	0.19
k_root	0.15	0.01	0.08	0.12	0.01	0.00



Morris

Physical interpretation:

- **Bending (f_1, w_tip)** is governed by the sandwich depth (t_core), face stiffness (E_1, t_ply) and mass (ρ_c). This is classic sandwich bending, $D \approx E_f\, t_f\, d^2/2$.
- **Torsion and higher modes (f_2, f_3, twist)** are governed by **core shear G_c**. This is the FSDT shear effect of Milestones 1–2: these modes would be wrong with k = 5/6 or with CLT.
- **Strength (RF_min)** is governed by **E_2 (matrix) and t_ply**, because the critical failure is matrix tension in the 90° ply.
- **Temperature** has a mid-level influence on every frequency. It is the main environmental confounder.
- **Ply misalignment (±2°) and core E_c** are negligible. They could be dropped from future DOEs, which is exactly what screening is for.
- σ (red dots) ≪ μ* everywhere, so the responses are almost linear and additive in these ranges.

9.2 First-order Sobol indices (from the 400 healthy LHS samples, bias-corrected)

	E_1	G_c	ρ_f	ρ_c	t_ply	t_core	T	others	**Σ S_i**
f_1	0.21	0.04	0.09	0.26	0.08	0.25	0.08	< 0.04	**1.03**
f_2	0.03	**0.56**	0.08	0.20	0.00	0.06	0.13	< 0.03	**1.08**
f_3	0.08	0.38	0.09	0.25	0.01	0.12	0.12	< 0.03	**1.08**
w_tip	0.15	0.00	0.01	0.04	0.37	0.35	0.05	< 0.01	**0.97**

- Σ S_i ≈ 1 confirms that the responses are **additive** (no significant interactions), in agreement with Morris.
- The ranking from both methods agrees.
- A raw (uncorrected) binned estimator gave Σ ≈ 1.27. The upward bias of (n_b − 1)/(n − 1) per factor is now removed analytically (`sobol_binned.m`).

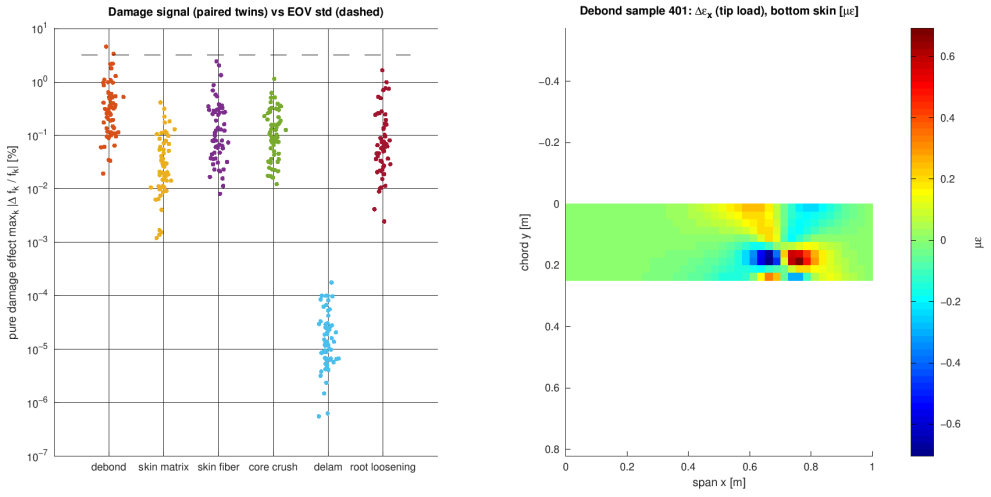
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10. Results — damage signal versus variability (the central SHM finding)

10.1 Global (frequency) signal

Damage class	Pure damage effect max_k |Δf_k/f_k|, median	90th percentile	Median signal / EOV std
Core debond	0.33 %	1.29 %	0.10
Skin matrix cracking	0.03 %	0.12 %	0.01
Skin fibre breakage	0.11 %	0.54 %	0.03
Core crush	0.11 %	0.39 %	0.03
Delamination (in-face)	≈ 0 (≤ $2 \cdot 10^{-6}$)	≈ 0	0.00
Root loosening	0.06 %	0.50 %	0.02
Healthy variability (EOV std)	**2.0 – 3.2 %** per mode		

Classical frequency-only novelty test (any $f_1 \ldots f_6$ outside the healthy mean ± 3σ): 0.5 % false alarms, but only **1 %** of debonds and **0 %** of all other damages detected.



Damage vs EOV

> **Finding 1.** Under realistic manufacturing scatter and temperature variation, natural frequencies alone **cannot** detect the damages of interest. The variability is 10–100 times larger than the damage signal. This is the well-known central problem of vibration-based SHM, and it is the scientific justification for the physics-informed features and AI of Milestones 4–5.

10.2 Local (strain-field) signal

The pure damage effect on the bottom-skin strain field under the 50 N tip load, max|Δε_x| / max|ε_x|:

Class	median	90th pct	**Median / EOV shape-variability**
Debond	0.40 %	1.54 %	**13×**
Skin matrix	0.89 %	3.37 %	**30×**
Skin fibre	1.37 %	30.0 %	**46×**
Core crush	0.22 %	0.56 %	≈ 7×
Root loosening	0.64 %	4.78 %	**22×**
Delamination	0.00 %	0.00 %	0

The **raw** strain variability from EOV is 4.3 % (median coefficient of variation per element), larger than most damage signals. However, EOV acts almost entirely as a **global scaling** of the field: stiffer material or a thicker core makes *all* strains smaller together. After removing **one scale factor per sample**, the remaining variability of the field **shape** is only **0.03 %**. Damage, by contrast, creates a **local pattern**, such as the dipole around the debond in the figure.

> **Finding 2 — the key design rule for Milestone 4.** Use **spatially normalised strain-field / mode-shape features**: ratios, residuals after global scaling, or curvature. They suppress environmental variability by about two orders of magnitude while keeping the local damage signature. This is where the physics-informed feature engineering earns its name.
>
> **Caveat.** These numbers are noise-free. Realistic strain-sensor noise (about 1 με, roughly 0.4 % of the peak strain here) will set the practical detection floor. Noise is added in Milestone 4 so its effect is quantified, not assumed.

10.3 Physical detectability limit: in-face delamination

A delamination *inside* a 0.5 mm carbon face changes the global response by ≤ $2 \cdot 10^{-6}$. The damage *is* applied: the section's shear correction factor changes from 0.095 to 0.120. But in a sandwich: - the **core** carries the transverse shear; - the delaminated sub-laminates still carry their full membrane load.

So bending stiffness is unchanged. This is a real physical fact, not a modelling bug. Such delaminations are found in practice with **ultrasonic / guided-wave** methods, not low-frequency vibration or strain.

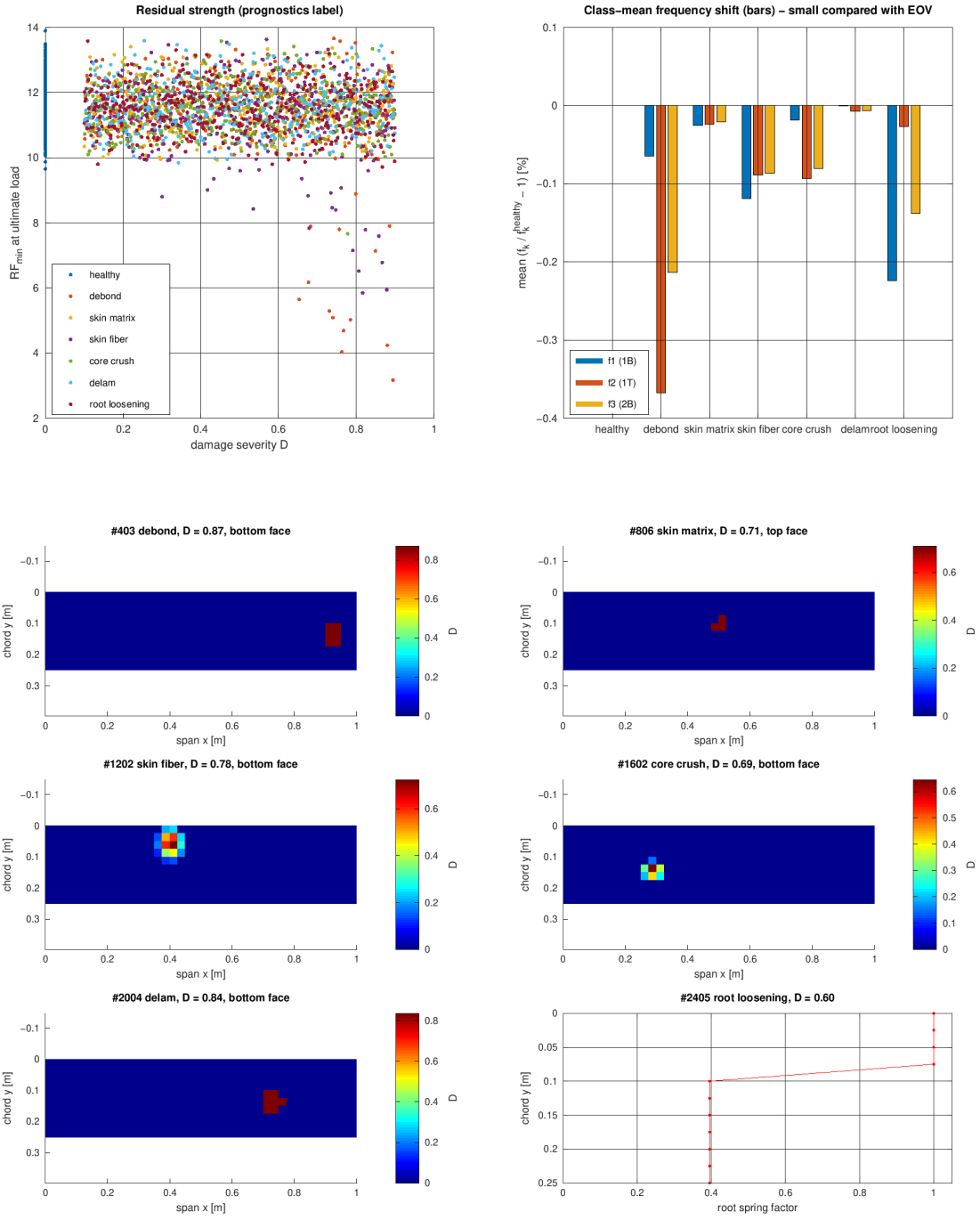
> **Finding 3.** The digital twin must **report it cannot see** this damage class rather than guess. Plan for Milestones 4–5: - keep the class in the database, honestly labelled; - evaluate the classifier's per-class probability of detection; - list guided-wave sensing as the recommended extension.
>
> Delaminations at the **face–core interface** are covered by the debond class, which *is* detectable.

10.4 Residual strength (prognostics label)

Class	RF_min mean ± std	worst sample
Healthy	11.58 ± 0.72	9.66
Debond	11.36 ± 1.31	**3.17**

Class	RF_min mean ± std	worst sample
Skin fibre	11.36 ± 1.08	5.85
Core crush	11.56 ± 0.75	7.67
Skin matrix	11.56 ± 0.73	9.93
Delamination	11.58 ± 0.74	9.85
Root loosening	11.54 ± 0.73	9.71

Severe debonds and fibre breakage can cut the reserve factor by **3×**. Strength loss, not stiffness loss, is what matters for safety, so RF is the **prognostics target** for Step 9 (remaining-useful-life estimation).



Gust sensitivity. The sharp-gust dynamic amplification factor is 1.686 for every sample. This is a consistency check, not missing physics. The sharp gust is defined relative to each sample's own

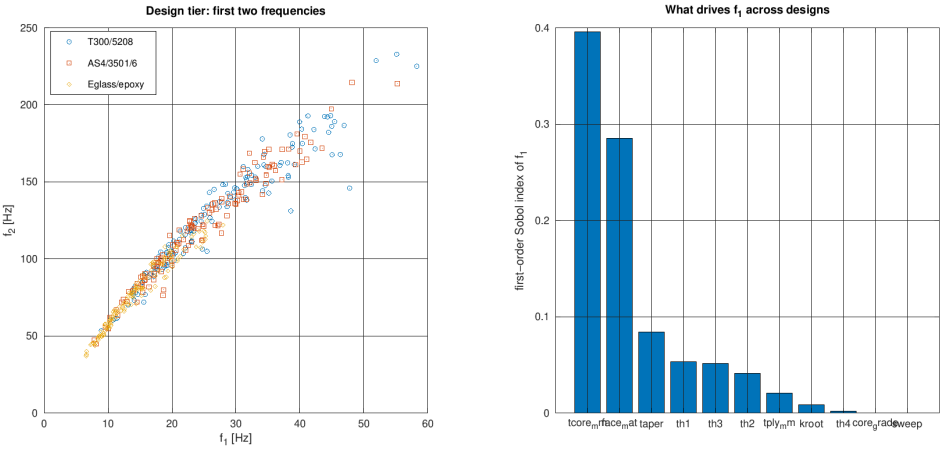
period ($t_g = T_1$), and the response is dominated by mode 1 with $\zeta_1 \approx 2$ %. The 1-cos shock-response spectrum value at $t_g/T_1 = 1$, $\zeta = 2$ % is ≈ 1.69, independent of stiffness.

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11. Results — design tier

Face material	n	f_1 range [Hz]	w_tip at 1 g [mm]	RF_min range
T300/5208	133	8.95 – 58.30	0.29 – 20.65	0.94 – 24.27
AS4/3501-6	134	7.83 – 55.27	0.38 – 22.40	1.69 – 25.36
E-glass/epoxy	133	6.57 – 28.00	1.13 – 38.13	0.77 – 16.42

- **f_1 first-order Sobol ranking:** core thickness (0.40) > face material (0.29) > taper (0.08) > θ_1 (0.05) > θ_3 (0.05).
- A few designs have RF < 1. These are genuinely under-designed wings. They are kept on purpose, so the surrogate also learns where the design space becomes unsafe.



Design tier

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12. Database contents (for Milestones 4–6)

Files are in `results/doe/`. The `.mat` files are MATLAB v7, readable in Python with `scipy.io.loadmat`. Large arrays are single precision.

Field	Size (SHM)	Content
`U`, `P`, `var_names`, `units`	2800 × 18	Inputs: unit-cube and physical values
`cls`, `classes`, `Dmap`, `root_factor`	2800, 2800 × 400, 2800 × 11	**Labels:** class, element-wise damage severity map, root spring factors
`freq`, `zeta`, `mode_type`, `mode_order`	2800 × 8	Modal data and automatic mode labels
`Wmodes`	451 × 8 × 2800	Mass-normalised w-mode shapes at every node

Field	Size (SHM)	Content
`Emodes`	400 × 3 × 8 × 2800	Bottom-skin modal strains (ε_x, ε_y, γ_xy) per element
`w_static`, `eps_bot`, `eps_top`	451 × 2 × N, 400 × 3 × 2 × N	Static fields for L1 (1 g) and L2 (tip load)
`tipw`, `twist`	N × 3 × 2, N × 2	Tip deflection LE / mid / TE, twist
`gam_L1`, `gam_L2`	N × 8	Modal forces, which regenerate any dynamic response
`RF`, `crit_elem`, `crit_hashin_mode`	N × 6	Residual strength at ultimate load
`gust`	N × 5	DAF, peak tip w / a, peak root strain, sharp-gust DAF
`ok`, `msg`, `time`, `qc`	N	Run status and quality checks

Also included: - `shm_scalars.csv` / `design_scalars.csv`: one row per sample with inputs and scalar outputs, for Excel / pandas. - `morris.mat`, `twins.mat`: the sensitivity and paired-twin studies.

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13. How to reproduce / verify in MATLAB

```
        >> setup_paths
        >> run_all_validation   % 56/56 checks (incl. new V10 mode-shape and V11 modal-solver checks)
        >> run_milestone3       % builds designs, runs all simulations (resumable), analysis
```

Runtime: about 1.6 h of serial CPU for all 3 200 simulations in Octave; MATLAB is usually faster. To parallelise: - set `USE_PARFOR = true` (Parallel Computing Toolbox); or - open several MATLAB windows and call `generate_database('shm', c, 4, 'results/doe')` with c = 1…4, one per window.

The run is check-pointed, so an interrupted run continues where it stopped.

Quick check without re-running everything: load the supplied `shm_scalars.csv` and compare a few rows. For example, sample 1 is healthy with $f_1 \approx 30.9$ Hz, and the per-class means are in Section 10.4.

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14. Summary

Step 3–4 goal (workflow diagram)	Delivered
Vary ply orientations, thickness ratios, materials	Design tier: 4 ply angles, t_f / t_c, 3 materials, core grade, taper, sweep
Load magnitudes & types	Linear superposition: unit 1 g, tip and gust responses plus modal forces (any n)
Boundary conditions	Elastic root fixity (log-uniform) and a root-loosening damage class

Step 3–4 goal (workflow diagram)	Delivered
Damage scenarios	6 physical damage classes, stratified, with location, size, severity, face
Batch high-fidelity simulation	3 200 FSDT runs, 0 failures, checked for equilibrium and eigen-residuals
Extract outputs	Frequencies, mode shapes, modal strains, static fields, strength, gust response
Comprehensive parameter space / large-scale data	Maximin-LHS (+13 % spacing, −13 % discrepancy), 228 MB database

Three scientific findings carry into Milestones 4–5: 1. Frequencies alone are swamped by EOV (signal/noise ≈ 0.01–0.1). 2. Spatially normalised strain-field features recover a 7–46× margin. 3. In-face delamination is physically invisible to this sensing approach. The twin must say so.

15. References

1. McKay, M.D., Beckman, R.J., Conover, W.J., "A comparison of three methods for selecting values of input variables," *Technometrics* 21, 1979.
2. Morris, M.D., Mitchell, T.J., "Exploratory designs for computational experiments," *J. Stat. Plan. Inference* 43, 1995.
3. Hickernell, F.J., "A generalized discrepancy and quadrature error bound," *Math. Comput.* 67, 1998.
4. Morris, M.D., "Factorial sampling plans for preliminary computational experiments," *Technometrics* 33, 1991.
5. Campolongo, F., Cariboni, J., Saltelli, A., "An effective screening design for sensitivity analysis of large models," *Environ. Model. Softw.* 22, 2007.
6. Saltelli, A. et al., *Global Sensitivity Analysis: The Primer*, Wiley, 2008.
7. Sohn, H., "Effects of environmental and operational variability on structural health monitoring," *Phil. Trans. R. Soc. A* 365, 2007.
8. Farrar, C.R., Worden, K., *Structural Health Monitoring: A Machine Learning Perspective*, Wiley, 2013.
9. Gibson, L.J., Ashby, M.F., *Cellular Solids*, 2nd ed., Cambridge, 1997.
10. Franklin, G.F., Powell, J.D., Workman, M., *Digital Control of Dynamic Systems*, 3rd ed. (first-order-hold discretisation).
11. Giurgiutiu, V., *Structural Health Monitoring with Piezoelectric Wafer Active Sensors*, 2nd ed., Academic Press, 2014 (guided waves for delamination).